Mathematics - Undergraduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the units digit in the standard decimal expansion of the number 7^25?
   (a) 1
   (b) 3
   (c) 5
   (d) 7

2. Find all zeros in the indicated finite field of the given polynomial with coefficients in that field. x^3
   (a) 1
   (b) 2
   (c) 2,3
   (d) 6

3. If f(z) is an analytic function that maps the entire finite complex plane into the real axis, then the i
   (a) the entire real axis
   (b) a point
   (c) a ray
   (d) an open finite interval

4. In the group G = {2, 4, 6, 8) under multiplication modulo 10, the identity element is
   (a) 6
   (b) 8
   (c) 4
   (d) 2

5. Let (Z_{10} , +, x) be the ring of integers modulo 10, and let S be the subset of Z_{10} represented by {0, 2, 4, 6, 8}. Which of the following statements is FALSE?
   (a) (S, +, x) is closed under addition modulo 10.
   (b) (S, +, x) is closed under multiplication modulo 10.
   (c) (S, +, x) has an identity under addition modulo 10.
   (d) (S, +, x) has no identity under multiplication modulo 10.

True / False (10 marks)

Answer True or False

6. True or False: Every nonzero free abelian group has a finite number of bases.
7. True or False: The symmetric group S_3 is cyclic, meaning it can be generated by a single element.
8. True or False: The homomorphic image of a cyclic group is always cyclic.
9. True or False: The probability that a randomly chosen real number x from [0, 3] is less than a ran
10. True or False: There are 6 structurally distinct Abelian groups of order 72.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Expand the following expression: $3(8x^2-2x+1)$.
12. Discuss the implications of linear independence among column vectors of a 5 x 5 real matrix M. Identify one condition that does not relate to the others and explain why.

End of Paper